

Faculty Course Assessment Report

Course: Data Structures and Algorithms

BSCS Program Outcomes

a.	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
b.	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
c.	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
d.	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
e.	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.
f.	Demonstrate use of team work, leadership skills, decision making and organization theory.
g.	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
h.	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.
i.	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.
j.	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

CS 323 Programing Fundamentals		Program Learning Outcomes									
		a	b	c	d	e	f	g	h	i	j
1	Understand the design of fundamental data structures as well as algorithms that operate on them.	X									
2	Get rigorous hands-on experience on implementing different data structures in a programming language.		X								
3	Summarize data structures according to their typical uses, strengths, and weaknesses for designing optimized computer-based systems.			X							

2	Get rigorous hands-on experience on implementing different data structures in a programming language.
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Head vise Grades		A	B	C	D	E	E	M	U	Avg.
Quiz 6		0	0	0	0	0	0	0	0	#####
Quiz 7		0	0	0	0	0	0	0	0	#####
Assignment 2		0	0	0	0	0	0	0	0	#####
Assignment 3		0	0	0	0	0	0	0	0	#####
Assignment 4		0	0	0	0	0	0	0	0	#####
Mid Term		0	0	0	0	0	0	0	0	#####
Final Term		0	0	0	0	0	0	0	0	#####
Avg.										#####

3	Summarize data structures according to their typical uses, strengths, and weaknesses for designing optimized computer-based systems.
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Head vise Grades		A	B	C	D	E	E	M	U	Avg.
Quiz 2		0	0	0	0	0	0	0	0	#####
Quiz 3		0	0	0	0	0	0	0	0	#####
Quiz 8		0	0	0	0	0	0	0	0	#####
Mid Term		0	0	0	0	0	0	0	0	#####
Final Term		0	0	0	0	0	0	0	0	#####
Avg.										#####

CS 323 Programing Fundamentals	Program Learning Outcomes		
	a	b	c
1. Understand the design of fundamental data structures as well as algorithms that operate on them.	#DIV/0!		
2. Get rigorous hands-on experience on implementing different data structures in a programming language.		#DIV/0!	
3. Summarize data structures according to their typical uses, strengths, and weaknesses for designing optimized computer-based systems.			#DIV/0!
EPAN (Average)	#DIV/0!	#DIV/0!	#DIV/0!
BS-CS (Metric Goal)	2.0	2.0	2.0
	#DIV/0!	#DIV/0!	#DIV/0!

Program Outcome Assessment	Assessing the student performance using Cohort Longitudinal Analysis, or CLA, the corresponding EPAN Average for program outcome a and b has (2.71, 2.51), which means the students demonstrated and had learned the outcomes a and b, and performed PROFICIENT. However, for 3rd program outcome, the corresponding EPAN Average of (0.57), which means the students performed at APPRENTICE level. However, it happens due to no submission of last assignment.
Student Feedback	According to the students, they understand the concept of Data Structures and have learned to implement different Data Structures as per requirement of Application. After attending this course they were very confident to take important decision for selecting appropriate data structures.
Reflection	More real word problem (case studies) should be discussed and solved in classes so that students can understand the unseen situations.
Proposed Actions for Course Improvement	Data base handling should be included in this course.